Hints: Linear algebra – 6. Linear algebra in combinatorics

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Fisher: the Gram matrix of the vectors $1_{A_i} \in \mathbb{R}^n$ is $G = \lambda J + D$ with $D = \operatorname{diag}(|A_i| - \lambda) \geqslant 0$. At most one $|A_i|$ equals λ, since two such sets would coincide. Show that $x^T G x = \lambda(\sum x_i)^2 + \sum d_i x_i^2 > 0$ for $x \neq 0$. Then G is invertible, so $m \leqslant n$.

Hint 2. Frankl–Wilson: use the polynomial method over $\mathbb{F}_p$. Let $a_i = 1_{A_i}$ and $f_i(x) = \prod_{l \in \mathbb{F}_p,\, l \neq \lambda}(x \cdot a_i - l)$. Then $f_i(a_j) = 0$ for $j \neq i$ and $f_i(a_i) \neq 0$.

Hint 3. Replace each x_k^2 by x_k (no change on 0/1 vectors). This gives multilinear polynomials of degree $\leqslant p-1$, still independent by the diagonal evaluation pattern. They live in a space of dimension $\binom{n}{p-1} + \cdots + \binom{n}{0}$.

Problem 2

Hint 1. Each biclique (L_t, R_t) gives the quadratic form $\ell_t(x) r_t(x)$, with $\ell_t = \sum_{i \in L_t} x_i$ and $r_t = \sum_{j \in R_t} x_j$. The partition says $\sum_{i<j} x_i x_j = \sum_{t=1}^{k} \ell_t r_t$.

Hint 2. The left side has matrix $\frac{1}{2}(J - E)$, with eigenvalue $-\frac{1}{2}$ of multiplicity $n-1$, so its negative index is $n-1$. Each $\ell_t r_t = \left(\frac{\ell_t + r_t}{2}\right)^2 - \left(\frac{\ell_t - r_t}{2}\right)^2$ has negative index $\leqslant 1$, and negative indices are subadditive. Hence $k \geqslant n-1$.

Hint 3. Equivalent elementary version: on the subspace $\{\sum x_i = 0,\ \ell_1 = \cdots = \ell_k = 0\}$, of dimension $\geqslant n-k-1$, the identity gives $\sum x_i^2 = 0$.

Problem 3

Hint 1. (IMC 2011.) Adding the same vector c to all v_i does not change sums of an even number of them. So you may translate so that the affine span of the v_i is a linear subspace W, of dimension $d \leqslant 2n-1$.

Hint 2. Lemma (induction on d): the vectors can be renumbered so that $v_1 + v_2,\ v_3 + v_4, \ldots, v_{2d-1} + v_{2d}$ form a basis of W. Pick a pair with $v_1 + v_2 \neq 0$ such that the remaining vectors still affinely span W, and pass to $W/\langle v_1 + v_2 \rangle$.

Hint 3. Let $w = v_1 + v_3 + \cdots + v_{2d-1} + v_{2d+1} + \cdots + v_{2n+d}$. This is $2n$ vectors, which is possible because $2n + d \leqslant 4n - 1$. Write $w = \sum_{i \in I}(v_{2i-1} + v_{2i})$. Adding this to w gives 0 as a sum of exactly one vector from each pair, plus the tail: $2n$ distinct vectors.

Problem 4

Hint 1. The vectors with odd indices $v_1, v_3, v_5, \ldots$ are pairwise orthogonal and nonzero, hence independent.

Hint 2. So $n \geqslant \lceil k/2 \rceil$. Conversely, $v_{2j-1} = v_{2j} = e_j$ works. Answer: $n = \lceil k/2 \rceil$.

Problem 5

Hint 1. The entry in row i, column j is $(i-1)n+j$. A choice of one entry in each row and column is $\{(i,\sigma(i))\}$. Its sum is $\sum_i (i-1)n + \sum_i \sigma(i)$, whatever σ is. The only possible value is $n(n^2+1)/2$.

Problem 6

(a)

Hint 1. If two such vectors have supports meeting in an odd number of coordinates, their inner product is a sum of an odd number of ± 1, so it is nonzero. For odd k this covers equal supports too, so all 2^k sign patterns on each support may be used.

Hint 2. You need 2^{k-1} supports of size k with pairwise odd intersections. Split $[2k]$ into pairs $\{1,2\},\dots,\{2k-1,2k\}$ and take the transversals: choose one element from each pair, with an even number of "second" choices. Two such transversals disagree in an even number of pairs, so they agree in an odd number. Total: $2^{k-1}\cdot 2^k$.

(b)

Hint 1. Upper bound: partition all $\binom{8}{4}2^4 = 1120$ vectors into 140 groups of 8 pairwise orthogonal vectors. A non-orthogonal family takes at most one vector from each group. Use a support S and its complement S^c, together with rows of a 4×4 Hadamard matrix.

Hint 2. In detail: the 16 sign vectors on S split into 4 orthogonal quadruples, namely $\pm$ the rows of H_4 and of $H_4\,\mathrm{diag}(1,1,1,-1)$. Vectors supported on S and on S^c are orthogonal. This gives $35\cdot 4$ octets.

Hint 3. Lower bound: let $A=\{1,2,3\}$ and $B=\{4,\dots,8\}$. Take all 0/1 vectors with two ones in A and two in B (30 of them), all vectors $1_A \pm e_b$, $b\in B$ (10), and the negatives of all of these. Check that every inner product is nonzero.

Problem 7

Hint 1. Choose J at random, putting each index in independently with probability $\frac{1}{2}$, and compute the expectation of the left side.

Hint 2. For $i\neq j$, the pair contributes a_{ij} with probability $\frac{1}{4}+\frac{1}{4}$. Since $a_{ii}=0$, the expectation is $\frac{1}{2}\sum a_{ij}$. Some J attains at least the mean.

Problem 8

Hint 1. For each i pick $u_i \in \operatorname{Im} A_i$ with $A_i u_i \neq 0$, e.g. $u_i = A_i x_i$ where $A_i^2 x_i \neq 0$. Show that $u_1,\dots,u_k$ are independent.

Hint 2. Apply A_i to $\sum c_j u_j = 0$. For $j\neq i$, $A_i u_j = A_i A_j x_j = 0$, so $c_i A_i u_i = 0$. Example with $k=n$: $A_i = E_{ii}$.

Problem 9

Hint 1. Let F be the family of sets with $s(I)=2019$ and V the set of values of s. Aim for $|F|\cdot|V| \leqslant 3^n$. Then $|F| \leqslant (3/1.8)^n < 1.7^n$.

Hint 2. For each value $v\in V$ fix one set J_v with $s(J_v)=v$. Map $(I,v)\mapsto 1_I - 1_{J_v} \in \{-1,0,1\}^n$.

Hint 3. This map is injective. From $d = 1_I - 1_{J_v}$ you recover $\sum d_i x_i = 2019 - v$, hence v, hence J_v and I.

Problem 10

Hint 1. If $|A| + |B| > p$, then for every x the sets A and $x - B$ intersect, so $A + B = \mathbb{Z}/p$. Otherwise assume $|A + B| \leqslant |A| + |B| - 2$. Take $C \supseteq A + B$ with $|C| = |A| + |B| - 2$.

Hint 2. $f(x, y) = \prod_{c \in C}(x + y - c)$ vanishes on $A \times B$ and has degree $|A| + |B| - 2$. The coefficient of $x^{|A|-1}y^{|B|-1}$ is $\binom{|A|+|B|-2}{|A|-1}$, which is nonzero mod p. The Combinatorial Nullstellensatz gives a contradiction.

Hint 3. Comparison: in $\mathbb{R}$, $\lambda(A + B) \geqslant \lambda(A) + \lambda(B)$ (place $\sup A + B$ and $A + \inf B$ side by side). In $\mathbb{Z}$ the same argument gives $|A| + |B| - 1$. In $\mathbb{Z}/p$ the wrap-around caps the bound at p, which is why an algebraic proof is needed.